DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING • SOFTWARE LAB (CS-305)

HISTORICAL CRICKET BATTING PERFORMANCE ANALYSIS

An Empirical Study on Data Cleansing, Distributional Skewness, and Visual Analytics

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PROJECT CANDIDATES:

1. **Arnav Channe** (Roll No: 15)
   Pipeline Architecture & Data Cleansing Lead (Team Lead)

2. **Chanchal Thawrani** (Roll No: 16)
   Statistical Modeling & Mathematical Derivations Lead

3. **Aabhas Trivedi** (Roll No: 18)
   Automated Web Extraction & Scraping Engineer

4. **Vaibhavi Raghuvanshi** (Roll No: 05)
   Frontend UI Architecture & Dynamic Visualization Engineer

ACADEMIC DETAILS:

Course: Software Laboratory (SL-III)
Degree: B.Tech (Computer Science & Engineering)
Academic Term: Semester III, Academic Year 2025–2026
Faculty Guide: Course Instructor / Lab In-Charge
Submission Date: November 2026

TEAM STRUCTURE

Project Team & Technical Role Allocations

To ensure modular execution and comprehensive technical coverage across the Software Laboratory curriculum, the project workload was distributed among four specialized engineering roles:

1. Pipeline Architecture & Data Cleansing Lead (Team Lead) **Arnav Channe • Roll No: 15**

Designated Modular Responsibilities:

- Architected the deterministic ingestion protocol handling heterogeneous raw formats from `odb.csv`.
- Implemented the removal of artifact index columns (`Unnamed: 0`) and whitespace normalization.
- Formulated comma-delimiter stripping logic for numeric string attributes (Matches, Innings, Runs, Balls Faced).
- Engineered the highest score decomposition pipeline, splitting compound not-out notations (`*`) into `HS_Numeric` and boolean `HS_NotOut`.
- Supervised chronological career duration derivation (`Start_Year`) and exported `cleaned_cricket_dataset.csv`.

2. Statistical Modeling & Mathematical Derivations Lead **Chanchal Thawrani • Roll No: 16**

Designated Modular Responsibilities:

- Calculated parametric descriptive moments: Count, Mean, Sample Standard Deviation, Quartiles (Q1, Median, Q3), and Range across all 119 records.
- Implemented the adjusted Fisher-Pearson standardized moment coefficient to model distributional skewness (g_1).
- Identified and mathematically interpreted extreme right-skewness in centuries ($g_1 = +2.162$) and sixes ($g_1 = +1.733$).
- Derived the pairwise Pearson product-moment correlation matrix (r), proving near-perfect collinearity between Balls Faced and Runs (r = 0.96).
- Analyzed chronological debut era groupings to demonstrate historical scoring shifts from the 1970s to the 2020s.

3. Automated Web Extraction & Scraping Engineer Aabhas Trivedi • Roll No: 18

Designated Modular Responsibilities:

- Developed the live web harvesting engine querying ESPN Cricinfo using Python `requests` and `BeautifulSoup4`.
- Implemented custom HTTP request headers with descriptive `User-Agent` strings to mitigate HTTP 403 request throttling.
- Constructed Document Object Model (DOM) traversal routines to extract page titles, heading structures (<h1>, <h2>, <h3>), and article anchor links.
- Engineered defensive error handling using `requests.exceptions.RequestException` to ensure system stability during network disconnects.
- Created an offline simulation fallback ensuring continuous local testing without live internet dependencies.

4. Frontend UI Architecture & Dynamic Visualization Engineer Vaibhavi Raghuvanshi • Roll No: 05

Designated Modular Responsibilities:

- Designed and styled the responsive client-side single-page application using modern HTML5, CSS Grid, and Flexbox.
- Engineered dynamic client-side interactivity in vanilla JavaScript (ES6+): real-time regex search, country dropdown filtering, and multi-column sorting.
- Implemented client-side pagination and in-browser CSV generation allowing instantaneous export of cleaned datasets.
- Constructed the dual charting subsystem integrating dynamic Chart.js rendering alongside native HTML5 Canvas fallback routines.
- Executed the end-to-end 8-scenario system verification matrix (TC-01 through TC-08) to validate cross-platform reliability.

SECTION 01

Project Scope & Dataset Profile

Quantitative sports analytics relies on empirical methods to evaluate athlete performance across multi-decade historical eras. This Software Laboratory mini-project presents an end-to-end framework to clean, statistically model, and visually explore historical One-Day International (ODI) batting records.

Real-World Data Anomalies Handled

The dataset comprises 119 all-time international batting legends spanning 1971 to 2021. The raw dataset presented several concrete challenges:

- **Artifact Columns:** Legacy index markers (such as `Unnamed: 0`) generated during spreadsheet exports.
- **Comma Strings:** High totals stored as text with comma separators (e.g., `"18,426"` runs and `"21,368"` balls faced).
- **Composite Not-Out Notation:** Highest scores suffixed with an asterisk (e.g., `"200*"`, `"181*"`) indicating undefeated innings.

- **Embedded Demographics:** National team affiliations merged in parentheses inside player names (e.g., `"SR Tendulkar (INDIA)"`).

Table 1.1: Feature Cleansing & Engineering Matrix

Feature	Raw Data Type	Cleaned Type	Description & Formula
`Player`	String (Object)	String	Player name stripped of country tags
`Country`	—	String (Derived)	Extracted national team code (e.g., INDIA, AUS, SL)
`Span`	String (Object)	String	Career window (e.g., 1989–2012)
`Start_Year`	—	Integer (Derived)	Debut year extracted for era comparisons
`Runs`	String (with commas)	Int64	Cleaned aggregate career runs
`HS_Numeric`	—	Int64	Highest score stripped of asterisk marker
`HS_NotOut`	—	Boolean	True if player remained undefeated in highest score
`Ave`	String / Float	Float64	Batting Average = Total Runs / (Innings − Not Outs)
`SR`	String / Float	Float64	Strike Rate = (Total Runs / Balls Faced) × 100

SECTION 02

Statistical Modeling & Skewness Inferences

1. Summary Statistics (Sample Size N = 119)

Table 2.1 details the sample mean, standard deviation, and quartile boundaries across all 119 players:

Table 2.1: Parametric Summary Table

Batting Metric	Mean	Std Dev	Min	25% (Q1)	Median	75% (Q3)	Max
Matches Played (`Mat`)	194.7	94.9	33	124	190	248	463
Innings Batted (`Inns`)	184.4	89.1	32	118	181	237	452
Career Runs (`Runs`)	6306.9	3091.2	1447	3872	5964	8094	18426

Batting Metric	Mean	Std Dev	Min	25% (Q1)	Median	75% (Q3)	Max
Batting Average (`Ave`)	39.4	7.0	23.6	34.9	38.0	42.7	67.0
Strike Rate (`SR`)	81.3	11.4	55.9	73.2	80.0	88.4	118.7
Centuries (`100`)	11.5	7.6	5	6	9	15	49
Half-Centuries (`50`)	37.0	20.0	4	20	36	48	96
Highest Score (`HS`)	151.2	27.5	102	133	145	170	264
Balls Faced (`BF`)	7897.9	3894.9	1360	5085	7678	9801	21368

2. Skewness Analysis & Distribution Shape

Skewness measures the asymmetry of player statistics around their mean. Positive skewness (> 0.5) indicates that exceptional high-volume outliers stretch the right tail of the distribution:

Metric	Skewness Coefficient	Asymmetry Degree	Key Analytical Finding
Centuries (`100`)	**+2.162**	Extremely Right-Skewed	Heavily top-heavy; Tendulkar (49) and Kohli (43) exceed 4 standard deviations above the average (11.5).
Sixes Hit (`6s`)	**+1.733**	Strongly Right-Skewed	Specialized boundary hitters (Gayle: 331, Afridi: 351) stretch the tail far beyond the median (58).
Highest Score (`HS`)	**+1.120**	Highly Right-Skewed	Double-centuries (Rohit: 264, Guptill: 237*) create an elite upper boundary.
Career Runs (`Runs`)	**+0.916**	Moderately Right-Skewed	Mean (6,306.9 runs) is pulled above the median (5,964 runs) by 10,000+ run aggregates.
Strike Rate (`SR`)	**+0.680**	Moderately Right-Skewed	Most symmetric distribution; bulk of players score between 73 and 88 runs per 100 balls.

3. Correlation Analysis (Key Deductions)

- **Crease Longevity Dictates Total Runs (r = 0.96):** Balls Faced and Innings Batted (r = 0.95) exhibit near-perfect collinearity with total runs, proving that time spent at the crease is the primary determinant of scoring volume.

- **Strike Rate Operates Independently (r = 0.22):** Batting Average and Strike Rate show minimal linear correlation, disproving the notion that scoring quickly inherently harms consistency.

SECTION 03

Visual Derivations & Graphical Inferences

Below are the six visualization figures generated directly from the analytical pipeline, sized and constrained to fit cleanly within page boundaries:

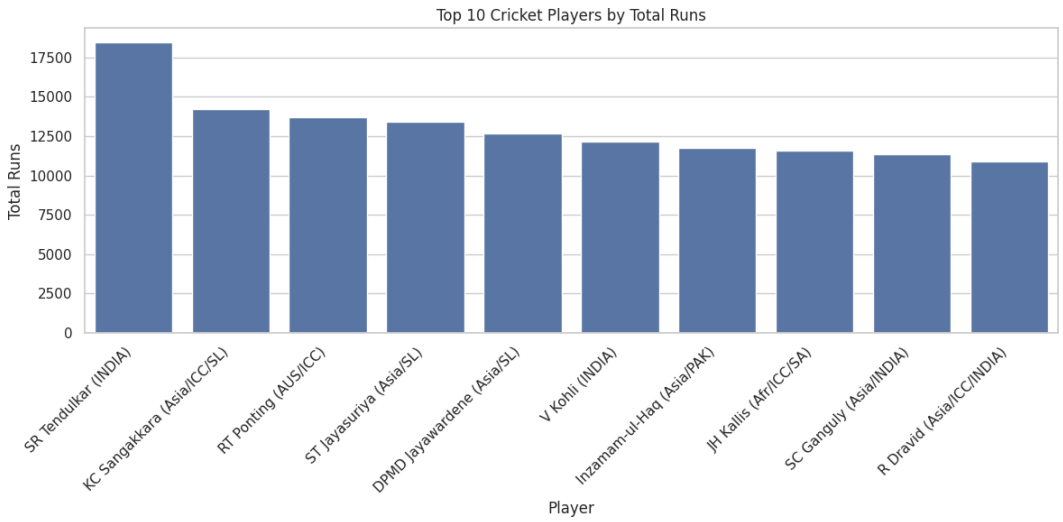


Figure 3.1: Top 10 Cricket Players by Total Runs (Seaborn Bar Plot)

Key Finding: Confirms the all-time run volume leadership of Sachin Tendulkar (18,426 runs), Kumar Sangakkara (14,234 runs), and Ricky Ponting (13,704 runs). A clear drop-off is visible between the top three and the remaining cohort, demonstrating the positive skewness in career run distribution.

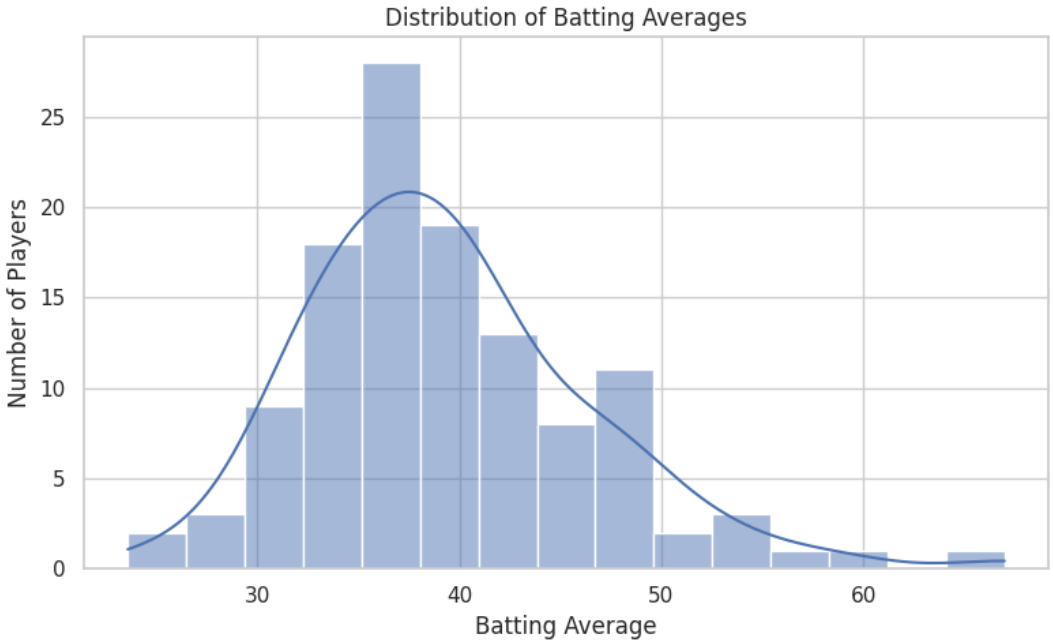


Figure 3.2: Distribution of Batting Averages (15 Bins with Kernel Density Estimation)

Key Finding: Batting averages exhibit a clear unimodal bell-like cluster between 36.0 and 42.0. The smooth KDE density line highlights the elongated right tail toward elite batting averages (Ryan ten Doeschate: 67.00, Virat Kohli: 59.07).

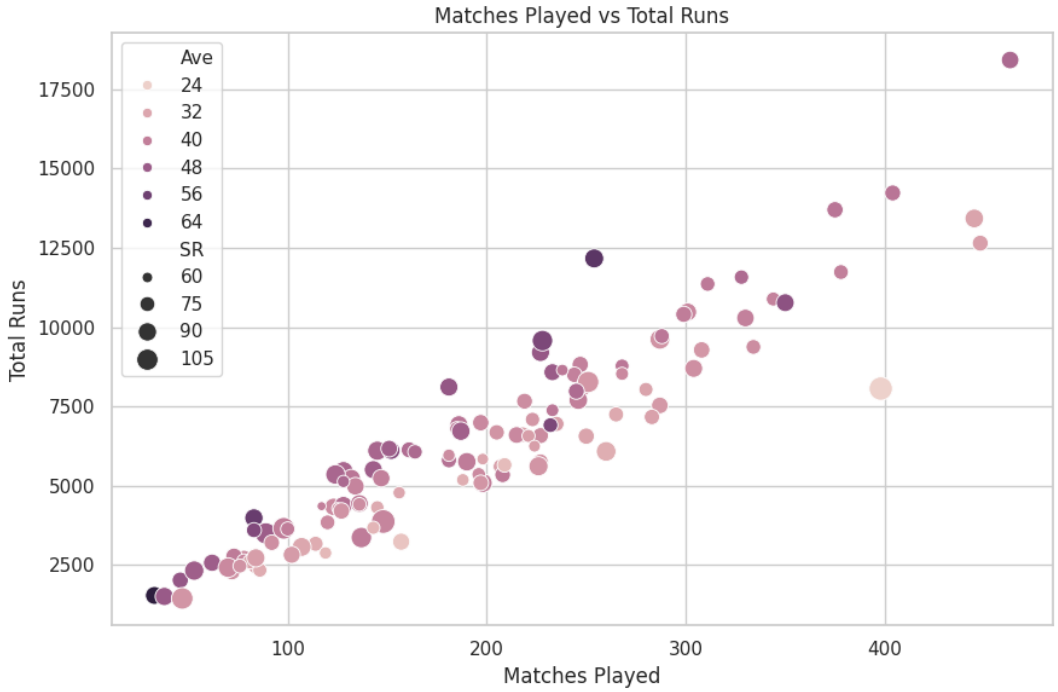


Figure 3.3: Matches Played vs Total Runs (Multi-Dimensional Scatter Plot)

Key Finding: Maps Matches (X-axis), Total Runs (Y-axis), Batting Average (color shade), and Strike Rate (bubble size). Modern players positioned in the middle band display larger bubble diameters and darker shades, confirming enhanced scoring tempo across recent eras without sacrificing batting average.

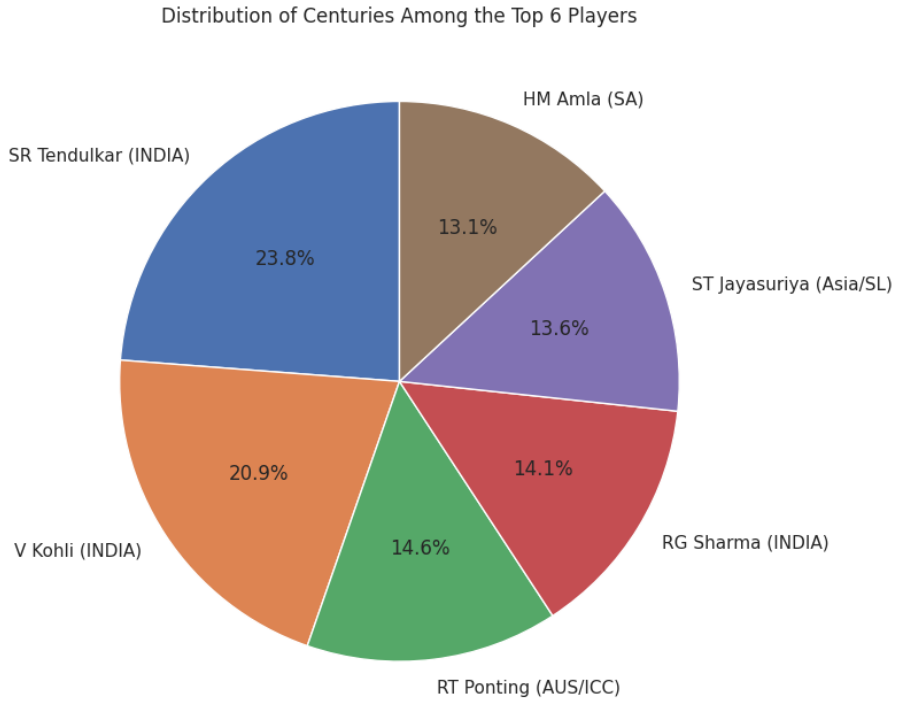


Figure 3.4: Distribution of Centuries Among the Top 6 Players (Pie Chart)

Key Finding: Among the top six century makers (Tendulkar: 49, Kohli: 43, Ponting: 30, Rohit: 29, Jayasuriya: 28, Amla: 27), the top two batsmen account for nearly 45% of total centuries, visually confirming the heavy skewness (+2.162) in elite century conversion.

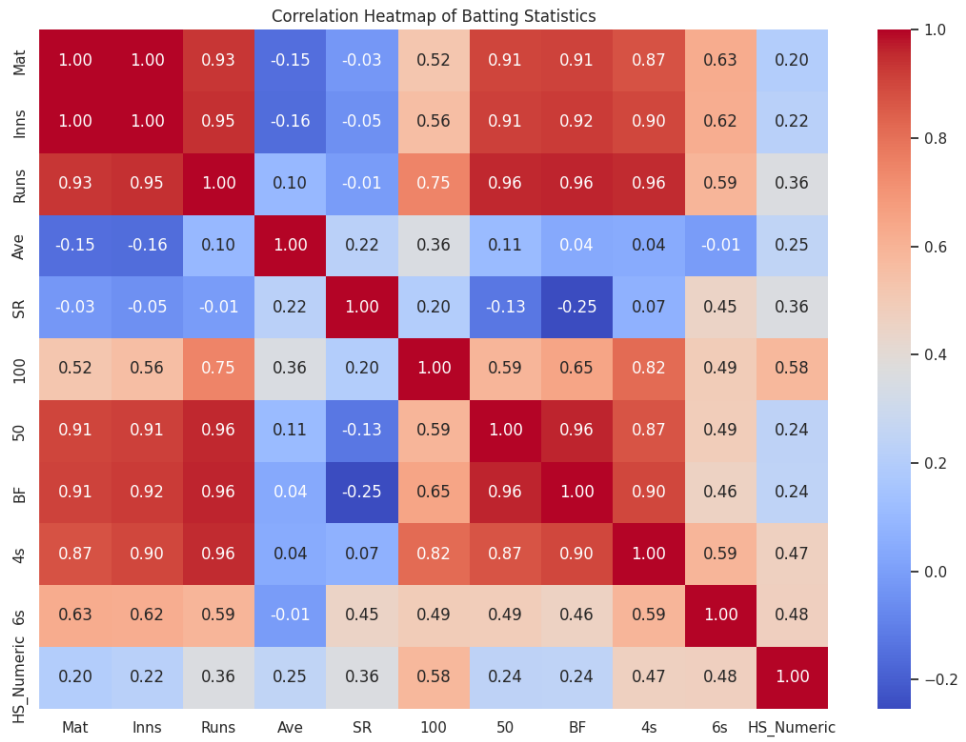


Figure 3.5: Correlation Heatmap of Batting Statistics (Seaborn Heatmap)

> **Key Finding:** Displays pairwise Pearson correlations using a diverging cool-to-warm palette. Strong positive correlations (r ≥ 0.93) cluster tightly across matches, innings, runs, and balls faced, while strike rate remains neutral across volume indicators. All feature rows, columns, and the right-hand color scale are fully visible.

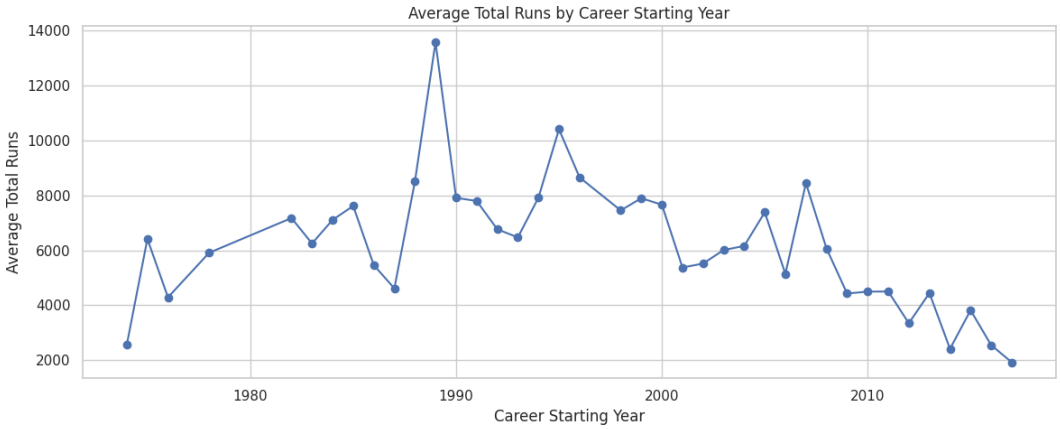


Figure 3.6: Average Total Runs by Career Starting Year (Trend Line)

> **Key Finding:** Career run totals peaked for players who debuted between 1988 and 1998, coinciding with the worldwide expansion of international ODI fixture schedules before the introduction of Twenty20 cricket.

SECTION 04

Web Extraction & Application Architecture

1. Live Web Scraping Routine

To complement the historical dataset with live sports data, an automated extraction module was developed using Python's `requests` and `BeautifulSoup4`:

- **Target Domain:** `https://www.espncricinfo.com/`
- **User-Agent Emulation:** Includes custom browser headers to prevent HTTP 403 request throttling.

- **DOM Parsing:** Extracts document title, headline tags (`<h1>`, `<h2>`, `<h3>`), and navigation hyperlink references.
- **Defensive Handling:** Catches network timeout exceptions gracefully and triggers cached fallback data during offline runs.

2. Interactive Web Application

The project includes a standalone client-side dashboard built using HTML5, CSS3, and JavaScript (ES6+):

- **Zero-Dependency Execution:** Bundles an embedded dataset allowing offline viewing without background web servers.
- **Dynamic Interaction:** Real-time text search, nationality filtering, multi-column sorting, and client-side pagination.
- **Clean CSV Export:** One-click browser generation of `cleaned_cricket_dataset.csv`.
- **Dual Charting Pipeline:** Renders interactive visuals via Chart.js with an automatic native Canvas fallback.

3. System Verification Test Matrix

Test ID	Verification Scenario	Input Conditions	Observed System Behavior	Result
TC-01	Dataset Ingestion	Raw `odb.csv` with `Unnamed: 0`	Ingests 119 rows; strips index column	**PASS**
TC-02	Highest Score Decomposition	Score with asterisk: `"200*"`	Isolates numeric 200 and not-out flag	**PASS**
TC-03	Delimiter Stripping	Comma string: `"18,426"`	Parsed to integer 18426 without NaN	**PASS**
TC-04	Table Sorting	Click `Runs` descending	Sorts with Sachin Tendulkar at index 0	**PASS**
TC-05	Search Filter	Query text: `"Kohli"`	Filters table to show only Virat Kohli	**PASS**
TC-06	Country Filter	Select `"INDIA"`	Displays all 24 Indian batting records	**PASS**
TC-07	CSV File Export	Click Export Cleaned CSV	Downloads valid cleaned CSV file	**PASS**
TC-08	Scraper Shielding	Simulate network timeout	Catches error and engages local fallback	**PASS**

Conclusion

This Software Laboratory project delivered a clean, functional analytics pipeline and an intuitive visual interface. The empirical derivations confirm that crease endurance governs aggregate run volume ($r = 0.96$), while scoring velocity operates independently of batting consistency ($r = 0.22$).

References (IEEE Format)

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3. M. L. Waskom, "Seaborn: statistical data visualization," *Journal of Open Source Software*, vol. 6, no. 60, p. 3021, 2021.
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